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Characterizing sustainability of green stormwater infrastructure technologies using life cycle assessment and life cycle costing: A systematic review

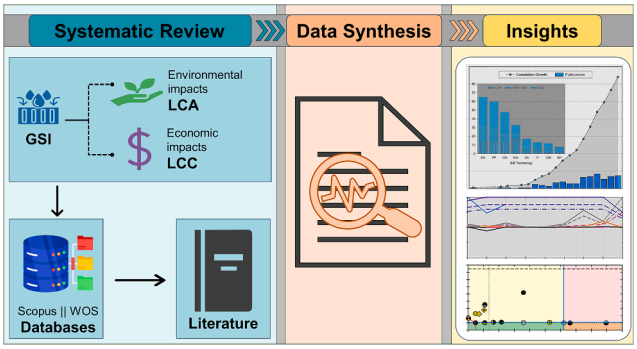
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HIGHLIGHTS

- 128 papers applying LCA and/or LCC to GSI were systematically reviewed.
- Reporting of life cycle impacts on a common per functional unit basis is recommended.
- Low economic impact may not indicate low environmental impact per functional unit.
- LCA and LCC can be coupled with other decision-making criteria for GSI sustainability.

GRAPHICAL ABSTRACT



ARTICLE INFO

Keywords:
Sustainable design
Life cycle assessment
Life cycle costing
Green stormwater infrastructure
Systematic literature review

ABSTRACT

Urban stormwater management often overlooks a holistic consideration of the functional, environmental, and economic aspects of green stormwater infrastructure (GSI). Life cycle assessment (LCA) and life cycle costing (LCC) tools are crucial for evaluating GSI; however, the variability in GSI implementation hinders generalizable conclusions for research, development, and deployment. To address this, we systematically reviewed 128 papers from Scopus and Web of Science that applied LCA and/or LCC to GSI, in alignment with Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines. We defined three research questions: (i) How have LCA and LCC been applied across different GSI technologies to assess sustainability performance? (ii)

Abbreviations: BR, Bioretention; CAC, Commercial activated carbon; CW, Constructed wetlands; FU, Functional unit; GR, Green roof; GSI, Green stormwater infrastructure; GWP, Global warming potential; ILCD, International Reference Life Cycle Data System; ISO, International Organization for Standardization; IT, Infiltration trench; IWSZ, Internal water storage zone; LCA, Life cycle assessment; LCC, Life cycle costing; LCI, Life cycle inventory; LCIA, Life cycle impact assessment; LID, Low-impact development; MS4, Municipal separate storm sewer system; NPV, Net present value; OD, Occurrence distribution; PP, Permeable pavement; PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses; PV, Present value; QMRA, Quantitative microbial risk assessment; QSD, Quantitative sustainable design; RH, Rainwater harvesting; RP, Retention pond; RTC, Real-time control; SBAC, Sludge-based activated carbon; SDG, Sustainable development goals; S-LCA, Social life cycle assessment; TRACI, Tool for Reduction and Assessment of Chemicals and Other Environmental Impacts; VS, Vegetated swale; WOS, Web of Science; WQV, Water quality volume.

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How do different design decisions influence the performance, environmental, and economic impacts of GSI technologies? (iii) How have LCA and LCC for GSI technologies been utilized to inform decision-making, and what opportunities exist to further their development? Our findings suggest that effective GSI design should prioritize and balance: (1) sufficient system size and water quality volume (WQV) to attenuate flow and reduce normalized sustainability impacts, (2) the inclusion of vegetation to enhance pollutant removal, and (3) extended lifespans to offset the initial environmental and economic impacts. Due to challenges we faced when comparing the sustainability performance of GSI of different scales, designs, and functionalities from various studies, our study emphasized the need to standardize the reporting of life cycle environmental and cost impacts of GSI on a common per-functional unit normalized basis (e.g., per unit of WQV treated). Trade-off analysis revealed that a GSI design with low economic impact does not automatically align with low environmental impact per unit of water quality volume treated. Ultimately, our study suggests that practitioners should apply outcomes from LCC and LCA, coupled with other decision-making criteria for GSI, to adequately balance stormwater management and minimize sustainability impacts. Based on the findings from our study, we examined the potential impact of stormwater policy on GSI adoption and identified gaps and emerging opportunities for enhancing the sustainability of GSI technologies.

1. Introduction

Imagine cities and urban areas where every drop of stormwater, whether precipitation or runoff, is harnessed to achieve sustainable environments and water resources that transform our world. Achieving this vision is possible through innovative research, development, and deployment efforts, which have led to the emergence, advancement, and advocacy of green stormwater infrastructure (GSI) technologies in recent decades (Chen et al., 2023; Davis et al., 2022; Dhakal and Chevalier, 2017; Shafique and Kim, 2017; Wang et al., 2023; WEF, 2012). GSI are innovative technologies that mimic the hydrological and ecological functions of the pre-development landscape to manage stormwater on-site by leveraging natural processes such as infiltration and evapotranspiration (Davis et al., 2022; WEF, 2012). Examples of GSI technologies include bioretention cells or rain gardens, permeable pavement, green roofs, and rainwater harvesting (Davis et al., 2022; Dhakal and Chevalier, 2017; WEF, 2012). Advancements in the GSI space contribute to the United Nations sustainable development goals (SDGs), particularly those focused on promoting clean water and sanitation, good health and well-being, sustainable and resilient infrastructure, sustainable cities and communities, and combating climate change and its impact (United Nations, 2015). These advancements in GSI technologies serve as a crucial stepping stone toward realizing the desired vision of cities of the future, where technology, social, economic, and environmental goals are balanced (Bashirpour Bonab et al., 2023; Chen et al., 2023; Javed et al., 2021; United Nations, 2015). With the increasing trends in urbanization, population growth, and climate change, GSI technologies have been cornerstones in national policies, strategies, and planning efforts to mitigate the anticipated increases in stormwater runoff (Dhakal and Chevalier, 2017). The integration of GSI in these capacities represents a significant shift towards creating more sustainable urban environments and resilient water infrastructure capable of withstanding these threats. Scientific evidence shows that GSI offer substantial benefits (affordability, reliability, sustainability, resilience) and can reduce environmental and economic impacts compared to traditional gray infrastructure (e.g., gutters, drains, and pipes). However, GSI technologies can be limited in their ability to manage stormwater in cases of heavy rainfall, especially during extreme storm events or in urban areas with large impervious areas (Davis et al., 2022; Dhakal and Chevalier, 2017). GSI technologies can provide safe and inclusive green public spaces, restore water-related ecosystem balance, and improve water quality by attenuating runoff and reducing water pollutants through natural processes, such as filtration and nutrient uptake (Davis et al., 2022; United Nations, 2015).

Despite the growing recognition of GSI, holistic research, development, and deployment for long-term sustainability remains a new and active area of research. Life cycle assessment (LCA) and life cycle costing (LCC) are sustainability assessment tools that respectively quantify the potential global environmental and economic impacts connected with a product or system (i.e., GSI technologies) throughout their life cycle (Asiedu and Gu, 1998; Finkbeiner et al., 2006; ISO 14040:2006, 2006; Li et al., 2022; Muralikrishna and Manickam, 2017; Rebitzer and Hunkeler, 2003). The application of LCA for environmental indicators and LCC for economic indicators provides decision-makers with a more comprehensive and holistic view of GSI technologies (Bixler et al., 2020, 2019; Hengen et al., 2016; Li et al., 2022; Wang et al., 2023). With LCA for GSI, the outcomes can be applied to analyze the contributions of the systems' life cycle phases (e.g., raw material extraction and processing, manufacturing or construction, operation or maintenance, end-of-life) to the overall environmental loads and/or compare different systems (De Haes and Van Rooijen, 2005; Muralikrishna and Manickam, 2017). As a result, LCA can help prioritize system improvements, inform strategic planning, and enhance the marketability of GSI (ISO 14040:2006, 2006; ISO 14044:2006, 2006; Muralikrishna and Manickam, 2017). LCC can assess all costs associated with GSI technologies across different life cycle phases, including design, production, usage, and disposal/recycling, thus enabling decision-makers to identify cost drivers and cost-effective design alternatives (Asiedu and Gu, 1998; Rebitzer and Hunkeler, 2003).

International Standards (ISO I4040 and ISO 14044) stipulate a four-step framework for conducting LCA studies (ISO 14040, 2006; ISO 14044, 2006): (1) goal and scope definition, (2) life cycle inventory analysis (LCI), (3) life cycle impact assessment (LCIA), and (4) life cycle interpretation. In step one, the goal and scope, the objective of the study, the system boundary, and the functional unit (i.e., the measure of system performance that enables fair comparison) are defined. In step two, LCI, all input/output data (foreground and background data) related to the GSI system are collected. Foreground data describes the specific system under study, and it is often gathered through modeling, field measurements, and experimentation (Muralikrishna and Manickam, 2017; van der Giesen et al., 2020). Background data is typically obtained from existing LCI databases (e.g., Ecoinvent) and literature, and it represents generic materials, energy, transport, and waste management processes (Muralikrishna and Manickam, 2017). In step three, LCIA, emissions (to land, water, air) and resources described in the LCI are translated into environmental impacts using characterization factors at the midpoint and/or endpoint levels (Huijbregts et al., 2017). In step four, interpretation, results are leveraged to provide a basis for conclusions and recommendations in alignment with the defined goal and scope (ISO 14040:2006, 2006; ISO 14044:2006, 2006). This phase includes uncertainty analysis, which is commonly accounted for using Monte Carlo simulation (Groen et al., 2017; Li et al., 2022; Vezzaro et al., 2013).

Synergistically, the implementation of LCC considers direct costs that are incurred by various actors in the life cycle, including suppliers (raw material costs), producers (production costs), users/consumers (operation and maintenance costs), and end-of-life actors (recycling and disposal costs). In addition, LCC can account for externalities, which are indirect or future costs such as environmental and social costs (e.g.,

pollution, resource depletion, and health impacts) (Rebitzer and Hunkeler, 2003). These externalities may be accounted for through regulatory measures such as anticipated carbon dioxide (CO_2) taxes to pay for carbon emissions from the GSI system, or renewable energy subsidies to encourage cleaner energy sources (Rebitzer and Hunkeler, 2003). Like LCA, uncertainty analysis can be conducted with LCC.

Over the past two decades, stormwater infrastructure research studies and literature reviews have focused on areas such as application (Shafique and Kim, 2017), treatment effectiveness (Collins et al., 2010), modeling (Elliott and Trowsdale, 2007), policy and governance (Li et al., 2019; Qiao et al., 2018), quantitative resilience (Shin et al., 2018), decision-making criteria (Bell et al., 2019), sustainability evaluation methods (Upadhyaya et al., 2014), sustainable urban infrastructure (Chen and Beck, 1997; Ferrer et al., 2018), low impact development (Shafique and Kim, 2017), LCA (Byrne et al., 2017b; Corominas et al., 2020, 2013; Loubet et al., 2014), carbon footprint (Kavehei et al., 2018), ecosystem services (Wang et al., 2023), and smart stormwater systems (Chen et al., 2023; Webber et al., 2022). While these studies encompass both gray and green infrastructure, to the authors' knowledge, there has been no systematic review that holistically considers the functional, environmental, and economic aspects specific to GSI, despite the growing popularity of GSI-focused LCA and LCC research. To this end, the objective of this systematic review was to examine the state, application, and opportunities for LCA and LCC methodologies in the context of GSI technologies. Three research questions were defined: *(i)* How have LCA and LCC been applied across different GSI technologies to assess their sustainability performance? *(ii)* How do different design decisions influence the performance, environmental, and economic impacts of GSI technologies? *(iii)* How have LCA and LCC for GSI technologies been utilized to inform decision-making, and what opportunities exist to further their development?

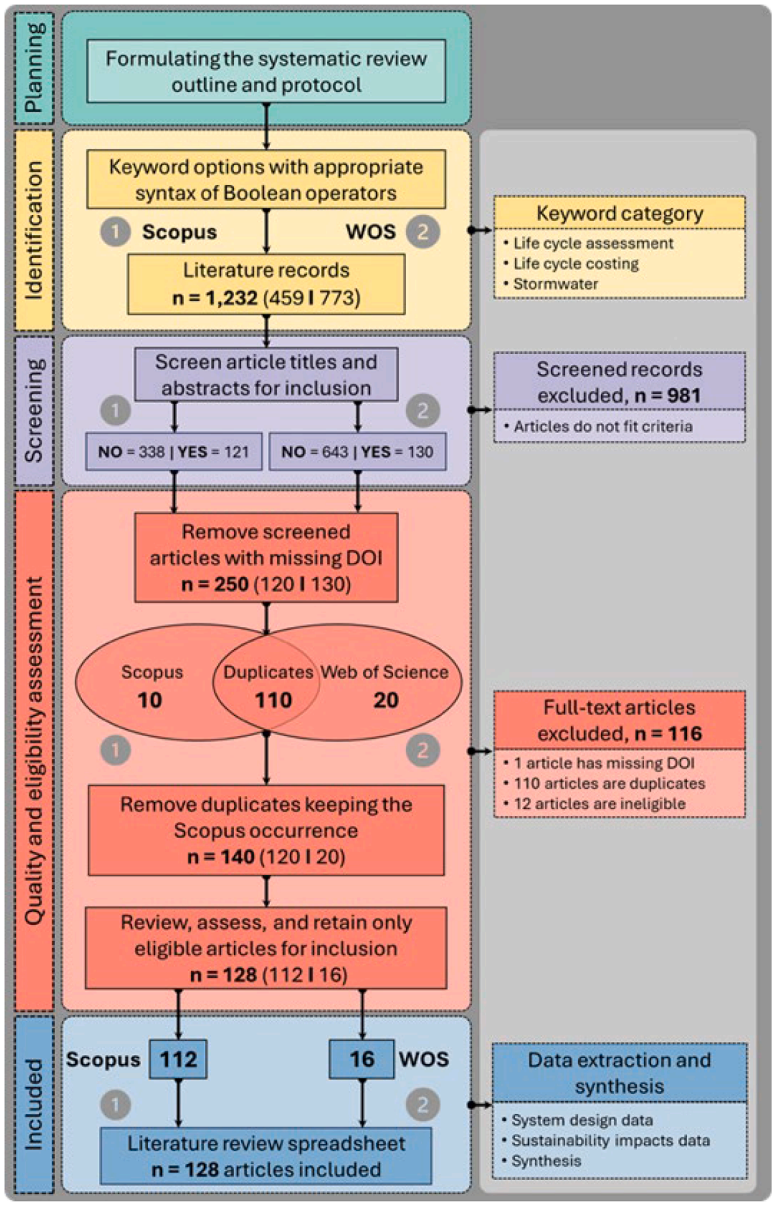


Fig. 1. Systematic literature review methodology, illustrating the process of retrieving studies from databases and applying inclusion and exclusion criteria. The circled numbers (1 and 2) represent the process flow for Scopus and Web of Science (WOS), respectively. The number of articles is defined as follows: n = total articles (Scopus articles | WOS articles).

2. Research methodology

2.1. Overview of systematic review

The peer-reviewed literature was systematically reviewed following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines and protocols – refer to Table S17 of the supplementary information (SI) for the detailed PRISMA checklist (Page et al., 2021; Varsha et al., 2024; Xiao and Watson, 2019). The systematic review process included steps of planning, identification, screening for inclusion, quality and eligibility assessment, and review (Fig. 1). The focus was exclusively on LCA and LCC studies related to GSI (e.g., bioretention cells, green roofs, wetlands, permeable pavements). GSI studies that focused on social life cycle assessment (S-LCA) for social impacts and quantitative microbial risk assessment (QMRA) for health impacts were not intentionally incorporated in this systematic literature review. Still, they would be included if they also conducted LCA and/or LCC. The literature was restricted to journal articles or articles-in-press that were available in English. To ensure relevance, we included only studies that (1) aligned with our objectives, research questions, and disciplinary scope and (2) were indexed in Scopus or Web of Science. We considered publications from 1950 through 2024; notably, the first applications of LCA/LCC to green stormwater infrastructure did not appear until 2003.

2.2. Electronic database search and evaluation

Electronic databases of Scopus and Web of Science (WOS) were comprehensively queried in October 2024 to gather all relevant literature. The search strategy employed an advanced query approach using a string of defined keyword combinations related to LCA, LCC, and stormwater (Table 1). Social assessment methods were not included in this study. For each database, the keyword options were formatted and enclosed within the appropriate syntax of Boolean operators (AND, OR) to facilitate the advanced query search (Table S1 of SI). The queried literature was exported as Microsoft Excel CSV files, which included the studies' metadata with authors, title, year of publication, source title, DOI, and abstract.

To screen for inclusion, the authors independently reviewed the article titles and/or abstracts of 1232 studies (459 from Scopus and 773 from Web of Science) and performed parallel assessments of the full-text articles when necessary. A study was included if it covered LCA and/or LCC of a GSI system. Hybrid technologies (i.e., coupled green-gray infrastructure) were included by default during electronic database search and author screening. A study was excluded if: (i) the full article was not present in English, (ii) it was a book chapter, (iii) it was a literature review, (iv) it was not an LCA and/or LCC of an actual GSI system, (v) it was an LCA and/or LCC study of a GSI system but more of a decision support tool or model without presenting LCA or LCC results. Discrepancies between reviewers were flagged and discussed until a consensus was reached for further analysis.

Database cleaning was performed using Python 3.12.7 (Python Software Foundation, 2024) in Visual Studio Code version 1.88.1. The main objective of this process was to create a database that consisted only of studies that met the inclusion criteria, free of missing metadata and duplicates. Once the cleaned database was complete, full articles were reviewed for inclusion based on the study's scope, following the same inclusion and exclusion criteria described above. The included articles were then exported to a new database and prepared for article reviews, data extraction, and synthesis.

Table 1
Electronic database search keywords.

Keyword Category	Keyword Search Options
Life cycle assessment	LCA, life cycle assessment, lifecycle assessment, lifecycle analysis, life cycle analysis, life-cycle analysis, life-cycle assessment
Life cycle costing	LCC, life cycle cost*, life-cycle cost*, lifecycle cost*
Stormwater	GSI, green infrastructure, green stormwater infrastructure, stormwater, storm water, drainage, LID, low impact development, low-impact development

* Wildcard character captures multiple word forms after the root word "cost" (e.g., costs, costing).

2.3. Data extraction and synthesis

The authors extracted data from eligible full-text articles using a structured Excel spreadsheet from the following overarching categories: article objectives, assessment tools, goal and scope, life cycle inventory, country of implementation, water and greenhouse gas direct emissions, impact assessment, uncertainty and sensitivity, interpretation, and key findings. To ensure accuracy and consistency, authors collaboratively reviewed all extracted data, discussing and resolving any discrepancies to reach a consensus. For all GSI technologies, LCA and LCC data were systematically extracted (Excel SI).

To better understand the influence of different design decisions on performance, environmental, and economic impacts, a deeper analysis of bioretention technologies was conducted, as it was the most studied GSI system in the reviewed literature. This process involved isolating articles that explicitly conducted LCA or LCC analyses for bioretention technologies. Data such as the design storm, impervious fraction, and drainage area were extracted to compute the design water quality volume (WQV) using the Schueler Simple Method (Equation S1 of SI) (Davis et al., 2022; WEF, 2012). WQV, a runoff volume defined by regulatory agencies, must be captured and treated, providing a basis for the design and sizing of bioretention and other GSI (Davis et al., 2022). For this study, the Schueler Simple Method was selected for WQV calculation due to its limited variable requirements and simple equation, making it the most straightforward method when calculating WQV from various bioretention studies included in the review. Because the Schueler Simple Method was developed for small catchments and assumes uniform soils, land cover, and hydraulic behavior, it does not capture spatial variability and may therefore underestimate the true WQV in more complex, mixed-use watersheds (Davis et al., 2022; WEF, 2012). In cases where the raw data necessary for WQV computation were not explicitly provided in the literature, values for parameters such as impervious fraction were derived using provided information (e.g., catchment/drainage area, impervious or pervious areas; Section S3 of SI). Furthermore, articles that explicitly reported the WQV in their bioretention designs were included in the analysis, even if they lacked sufficient data for calculating WQV, except for one design scenario where the reported WQV was identified as an outlier due to difference of 2 orders of magnitude compared to the other scenarios (Hengen et al., 2016). Other key design parameters extracted included bioretention system surface area, catchment characteristics, filter media composition, design lifespan, target stormwater pollutants, and vegetation types. Functional units (FUs) from the articles were grouped into broad categories, including water treatment, water storage, flood control, drainage area, impervious area, surface area, and water quality volume. Aggregated LCI data for bioretention (encompassing construction, operation, and end-of-life phases) were normalized to the WQV for each design scenario to enable comparison. Extracted life cycle impacts were categorized into three assessment methods, specifically TRACI (Tool for Reduction and Assessment of Chemicals and other Environmental Impacts), ReCiPe, or others, e.g., ILCD (International Reference Life Cycle Data System), CML (Centrum Milieukunde Leiden). Impact categories with two or fewer missing data points were analyzed further. Life cycle costs were also normalized to the WQV for comparative trade-off analysis and synthesis (Section S4 of SI).

3. Results and discussion

3.1. Systematic literature review database: selected studies and literature trends

The systematic literature review process yielded a pool of 1232 search hits from Scopus and Web of Science, ultimately narrowing down to 128 eligible articles (112 from Scopus and 16 unique additions from Web of Science) published between 2003 and 2024 (Fig. 2). The final 128 articles were reviewed and analyzed, revealing that 38 % (48 articles) focused on LCC studies, 34 % (44 articles) on LCA studies, and 28 % (36 articles) employed a combination of LCA and LCC. Most of these studies were published in water and environment-focused journals, including the Journal of Cleaner Production (20 articles), Science of the Total Environment (11), Journal of Environmental Management (10), Water Research (6), Resources, Conservation and Recycling (6), and Water (5) (Table S3 of SI). Geographically, China and the USA together accounted for over half (54 %) of the total literature studies — 28 % and 26 %, respectively (Figure S1 and Table S4 of SI). This dominance may be related to policy shifts toward sustainability and stormwater management, public preference for GSI, and a higher willingness to invest in GSI development and implementation (Ma et al., 2020; Meng and Hsu, 2019; R. Wang et al., 2022).

Although the annual number of eligible publications fluctuated in the past two decades, the cumulative trend showed consistent growth (Fig. 2). The past five years (2019–2024) demonstrated significant activity, with the year 2021 marking a peak, accounting for 13 % (17 articles) of the total publications. The growth coincides with the introduction and adoption of the United Nations Sustainable Development Goals (SDGs) (United Nations, 2015). Out of 30 GSI technologies, bioretention and permeable pavement dominated the literature, with each included in over 50 published articles (Fig. 2 inset). This prevalence demonstrates their widespread application in urban stormwater management, particularly in areas such as parking lots, sidewalks, and roadways. Sustainability assessment tools used in the reviewed literature varied by GSI. Bioretention and permeable pavement sustainability assessments were predominantly LCC (52 %), whereas rainwater harvesting assessments leaned toward LCA (47 %). Retention ponds and constructed wetlands were the most frequently evaluated GSI using both LCA and LCC tools (75 % for retention ponds, 58 % for constructed wetlands), thus suggesting a stronger interest in joint environmental-economic impacts (Table 2, Excel SI). Relative to expectations under independence, LCA and combined LCA+LCC assessments were conducted more than expected for rainwater harvesting and retention ponds, respectively (Section S2 of SI and Excel SI).

3.2. Application of LCA and LCC in evaluating the sustainability performance of GSI

Like other studies (Knapik, 2022), the results showed that different lexicons were used in studies to define similar GSI technologies. For example, rain gardens are sometimes referred to as biofilters or bioretention, and permeable pavements as porous pavements. To address the inconsistency in GSI terminology, 30 GSI technology categories were established based on every GSI found in the 128 studies (Excel SI). Similarly, the sustainability tools applied (LCA, LCC, LCA + LCC) and the corresponding metric categories (e.g., system boundary) and sub-metrics (e.g., construction) were used to further classify these GSI technologies (Excel SI). Results reported here only cover the top three sub-metrics, ranked by total occurrence, within each sustainability

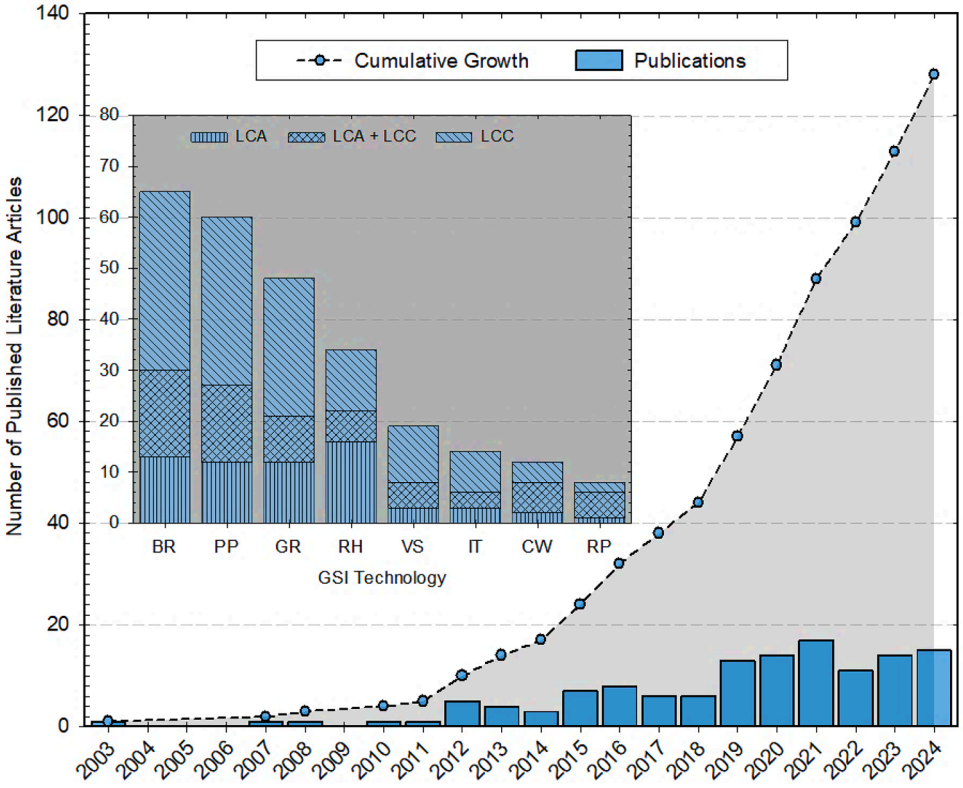


Fig. 2. Trends in GSI-related published literature applying LCA and/or LCC over the past two decades (2003 to 2024). The main chart illustrates the cumulative growth of reviewed articles (dotted line with circles) alongside annual publications (blue bars). The inset stacked bar chart categorizes the articles based on sustainability assessment tools (LCA, LCC, and LCA+LCC) used to evaluate selected GSI technologies, including bioretention (BR), permeable pavement (PP), green roof (GR), rainwater harvesting (RH), vegetated swale (VS), infiltration trench (IT), constructed wetlands (CW), and retention ponds (RP). Only GSI technologies studied in more than 8 articles were included in the inset stacked bar chart. Comprehensive details on these 8 GSI technologies and their descriptions are provided in Table S2-A of the SI.

Table 2

Occurrences of selected LCA and LCC metrics categorized by studied GSI (more details in Excel SI). Rows represent different GSIs, and columns represent LCA and LCC metric categories. A more filled occurrence distribution indicator (pie chart-like symbol) for a metric indicates its greater application to the GSI relative to other metrics in the same category. For example, more bioretention LCA studies included the construction phase compared to the end-of-life (EoL) phase, with numbers denoting individual occurrences. Acronyms include O&M (operation and maintenance), EoL (end of life), WQV (water quality volume), TSS (total suspended solids), EIO-LCA (economic input-output LCA), NPV (net present value), PV (present value), and CBR (cost-benefit ratio).

	Assessment Tool			System Boundary						Functional Unit			Direct Emissions			LCIA Tool			LCIA Categories			LCC Metric			Discount Rate			Uncertainty & Sensitivity		
				LCA			LCC																							
GSI Technology (Total Occurrences)	LCA	LCC	LCA + LCC	Construction	O & M	End of Life	Capital costs	O & M costs	EoL costs	WQV	Drainage area	System size	TSS	Total Phosphorus	Total Nitrogen	TRACI	ReCiPe	EIO-LCA	Acidification	Eutrophication	Global Warming	NPV	PV	CBR	2%	3%	5%	Uncertainty	Sensitivity	Both analyses
Bioretention (65)	13	34	18	31	29	14	49	47	13	8	6	4	12	13	9	8	13	2	21	22	30	11	21	6	4	5	24	20	23	10
Permeable Pavement (60)	12	3	17	29	25	13	46	44	14	5	5	4	7	7	4	4	13	3	18	19	27	11	20	5	4	3	21	13	20	5
Green Roof (48)	12	26	10	22	18	12	35	34	6	1	4	11	5	5	3	1	6	2	15	15	20	6	19	5	1	3	16	9	19	5
Rainwater Harvesting (34)	16	11	7	23	22	9	17	18	3	2	1	1	3	5	6	7	5	4	9	12	20	5	8	3	1	3	6	7	16	3
Vegetated Swale (19)	3	10	6	9	9	5	15	15	4	3	0	2	3	3	2	2	4	1	6	7	9	2	5	3	0	0	8	4	8	2
Infiltration Trench (14)	3	7	4	7	5	4	11	11	6	3	0	0	1	0	0	1	3	1	6	5	6	2	5	1	0	2	5	2	7	1
Constructed Wetlands (12)	2	3	7	9	9	3	9	9	1	2	1	0	4	4	4	2	6	1	5	7	8	3	1	2	0	1	4	5	8	2
Retention Ponds (8)	1	1	6	7	7	4	7	7	2	1	0	1	3	2	1	4	3	1	5	6	6	2	3	1	0	0	4	3	6	2

Scale showing the class limits for the GSI occurrence distribution (OD) indicators in the table

0%	>0 – 10%	>10 – 20%	>20 – 30%	>30 – 40%	>40 – 50%	>50 – 60%	>60 – 70%	>70 – 80%	>80 – 90%	>90 – 100%

metric category for the eight most frequently studied GSI technologies, which together appear in 124 of the 128 reviewed articles (Table 2).

The system boundary and functional unit play a critical role in defining the goal and scope and informing findings of any LCA and/or LCC study (ISO 14040:2006, 2006; ISO 14044:2006, 2006; Rebitzer and Hunkeler, 2003). The LCA system boundary results indicated that the construction phase (encompassing raw material extraction, processing, installation, and transportation) was the most extensively considered life cycle phase across the top eight GSI technologies, with an occurrence distribution (OD) of >90–100 % (Table 2). We define OD as the frequency at which a specific GSI technology was studied in the literature, measured based on a given sustainability methodology and metric category (e.g., LCA and functional unit, respectively) – and since a study can examine multiple GSI technologies, the total occurrences may exceed the total number of literature articles in this review. The construction phase was followed by the operations and maintenance (O&M) phase, with a minimum OD of >70–80 %, and finally, the end-of-life (EoL) phase, which includes decommissioning, disposal, recycling, or reuse, with an OD of at least >30–40 % across the top eight GSI technologies. The LCC system boundary showed a similar pattern, albeit with distinct ODs (Table 2). While some GSI studies include the construction phase due to the expected significant environmental and economic impacts, its frequent consideration in LCA and LCC system boundaries is also driven by the availability of inventory data, unlike the O&M and EoL phases (Abdeljaber et al., 2022; Byrne et al., 2017a; Wang et al., 2013). Construction data can often easily be sourced from design documents, manufacturers, contractors, and cost estimation tools like RSMeans (Byrne et al., 2017a; RSMeans, 2012; C. Xu et al., 2021), and data related to transportation distances can be collected freely using digital maps to compute transport allocations (O'Sullivan et al., 2015; Vaz and Ghisi, 2024). In contrast, GSI O&M data is highly variable and depends on external factors (e.g., stormwater runoff and treatment depend on rainfall intensity), thus necessitating continuous monitoring, which is complex and costly (Davis et al., 2022; Rieck et al., 2022; Vineyard et al., 2015; Wang et al., 2013). In practice, GSI technologies are commonly and mistakenly considered as "install-and-forget" technologies, a misconception that has resulted in zero to minimal maintenance, lack of O&M inventory data collection, unclear management roles, and nonexistent maintenance budgets (Knapik, 2022; Knapik et al., 2024). The lack of O&M data prevents tracking of crucial performance metrics such as flow attenuation, pollutant removal efficiencies, and carbon sequestration, which in turn obscures the true economic and environmental trade-offs of GSI. Increasingly, studies have excluded the EoL phase due to its reported insignificant impacts relative to the other phases (Bixler et al., 2019; Flynn and Traver, 2013; Jeong et al., 2016). Moreover, as first-generation installations of GSI now approach their design service lives, there is substantial uncertainty regarding EoL performance, standard procedures for EoL, data collection, and costs (Rizzo et al., 2023). This further contributes to missing or unreliable EoL inventory data and the exclusion of the EoL phase (Abdeljaber et al., 2022).

The most frequently used functional units were related to WQV (25 occurrences), system size defined by volume and/or area (23), and drainage area (17) across GSI technologies. Contrary to the ISO guidelines, some LCA studies did not explicitly define or specify a measurable FU (21 occurrences). Studies applied FUs based on the specific performance and design objectives of GSI technologies. For example, green roofs predominantly used system size (surface area) as the FU (Table 2) in alignment with design objectives focused on retaining stormwater and facilitating evapotranspiration (Davis et al., 2022). In contrast,

bioretention technologies and permeable pavements rely on WQV and drainage area as dominant FUs, reflecting their emphasis on stormwater control, treatment, and infiltration. Because FUs are inherently tied to the specific performance characteristics of each system, some LCA studies employed two or more FUs for GSI technologies designed for multiple functions (Byrne et al., 2017a; Loubet et al., 2014; Teston et al., 2024; Xu and Zhang, 2019). These observed variations in FU choices not only inform how technologies are designed but also affect how the sustainability impacts of GSI technologies are interpreted under uncertainty. For instance, selecting a volume-based FU (e.g., WQV) may give precedence to designs that maximize infiltration, flood control, and treatment per unit volume, while a user-based FU (e.g., water consumption per capita, ecosystem services) may emphasize multifunctional designs that also incorporate biodiversity, recreation, and social benefits (Teston et al., 2024).

GSI sustainability assessments had other notable trends. For example, total suspended solids, total phosphorus, total nitrogen, zinc, and copper were among the most commonly assessed water quality direct emissions, with bioretention technologies and permeable pavements exhibiting the highest representation, likely due to their pollutant removal capabilities (Davis et al., 2022; WEF, 2012). Meanwhile, greenhouse gas (GHG) direct emissions were almost never included, sometimes due to their perceived negligible impact (Kavehei et al., 2018). Although GHG emissions would factor into some environmental impact categories (i.e., global warming potential or climate change), those GHG estimates are often based on background data alone, and LCI database estimates are often limited to generalized emission factor approaches. Yet, GHG direct emissions (both positive as production and negative as removal) should be considered, particularly within GSI that harness biological removal mechanisms (e.g., denitrification for nitrate removal and nitrous oxide production, plant uptake for CO_2 removal and biomass production) to ensure an accurate accounting of their climate change impacts. Considering LCIA, environmental impacts were typically evaluated using LCIA tools such as TRACI, ReCiPe, and EIO-LCA, with SimaPro being the most widely used LCA software. The most frequently analyzed impact categories included global warming potential, eutrophication, and acidification, indicating a strong research focus on climate change mitigation and water quality improvements. From an economic perspective, LCC assessments primarily relied on present value (PV), sometimes referred to as net present value (NPV), and cost-benefit ratio metrics. To account for the time value of money, a 5 % discount rate was frequently adopted (followed by 3 % and 2 %; Table 2), reflecting varying assumptions about long-term cost implications. Although both uncertainty and sensitivity analyses are essential for improving the robustness of GSI LCA/LCC results, relatively few studies incorporated both methodologies simultaneously. This suggests the need for the application of these analyses to better capture variations in dynamic GSI technologies amidst data limitations and environmental

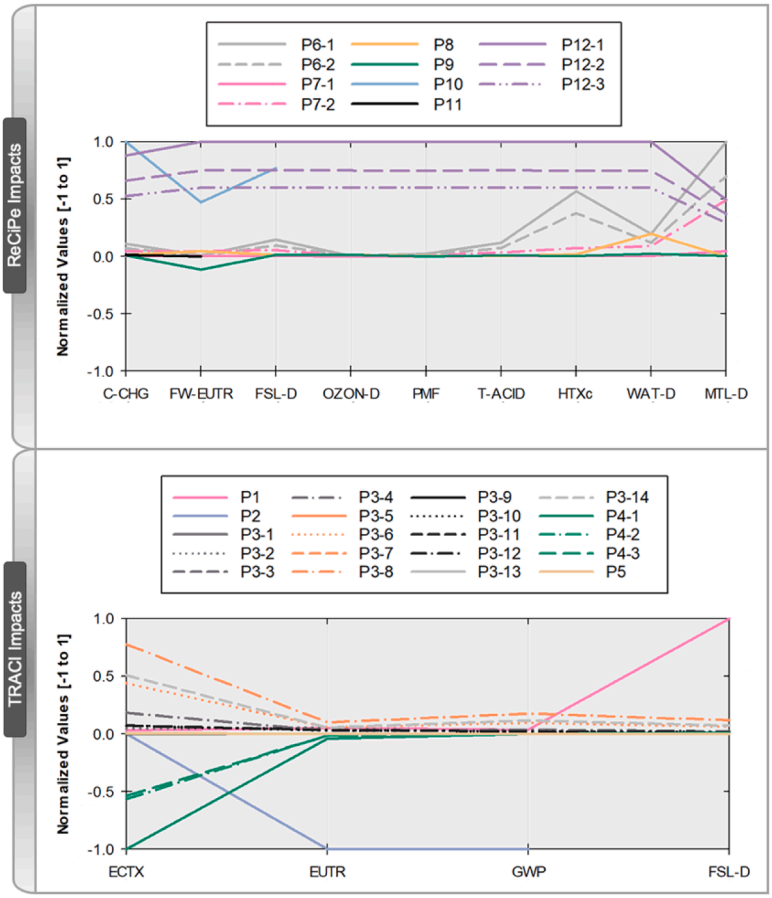


Fig. 3. Parallel coordinate plots comparing environmental impacts of 31 bioretention design scenarios. ReCiPe (top panel) and TRACI (bottom panel) life cycle midpoint environmental impacts are normalized to water quality volume (Section S3 of SI). ReCiPe impact categories include: C-CHG = climate change, FW-EUTR = freshwater eutrophication, FSL-D = fossil depletion, OZON-D = ozone depletion, PMF = particulate matter formation, T-ACID = terrestrial acidification, HTXc = human toxicity, WAT-D = water depletion, and MTL-D = metal depletion. TRACI impact categories include: ECTX = ecotoxicity, EUTR = eutrophication, GWP = global warming potential (climate change). Design names (e.g., P1, P6–1, P6–2) indicate the original paper from which each scenario was extracted, with numeric extensions denoting multiple scenarios in the same paper (Table 3). To enable direct comparison on a uniform scale (y-axis from −1 to 1), each impact metric (normalized to WQV) was divided by the highest value within its category. A value of 1 thus indicates the bioretention technology with the highest impact for that category. A negative value suggests potential credits or net benefits in that category.

stressors.

3.3. Influence of design decisions on the performance, environmental, and economic impacts of GSI technologies – a case study of bioretention technologies

The design and performance of GSI, such as bioretention, are shaped by a blend of design decisions, technology characteristics, and contextual factors (Davis et al., 2022; WEF, 2012). Moreover, subjective methodological choices like setting system boundaries, selecting the FU, and design and modeling assumptions are also critical in assessing performance and life cycle impacts of GSI technologies. To explore these differences, parallel coordinate plots were used to understand how different bioretention design decisions influence environmental impacts (Fig. 3 and Figures S2 and S3 of the SI). In this analysis, each bioretention scenario extracted from the reviewed literature was treated as a distinct design scenario named after the corresponding paper ID (Table 3). Descriptive analysis of bioretention design basis variables from reviewed literature showed that most bioretention studies represent small to moderate urban retrofits (i.e., the systems are modest in size), with notable outliers (Table 3, Section S3 of the SI, and Excel SI). The bioretention median values showed that the available storage or system sizes for the bioretention technologies in the reviewed studies were sufficient to capture the first inch of stormwater runoff and support bioretention natural processes, thus being consistent with the water quality design standards for highly impervious sites (Davis et al., 2022; WEF, 2012).

The absolute environmental impacts were normalized to water quality volume (WQV) to enable a standardized comparison of different designs. The calculated WQV used for this normalization process for all bioretention technologies can be seen in Table 3 or Section S3 of the SI. Bioretention scenarios evaluated using the ReCiPe methodology (P6 through P12) were designed with lifespans of 30 years (P6–1, P6–2, P7–1, P7–2, P9, P10, P11, P12–1), 40 years (P12–2), and 50 years (P12–3) (Table 3 and Table S5 of SI, Excel SI for descriptive analyses). Among the bioretention technologies, six were designed without vegetation (P6–1, P6–2, P7–1, P7–2, P8, P9), while five incorporated vegetation (P10, P11, P12–1, P12–2, P12–3). Bioretention scenarios evaluated with the TRACI methodology (P1 through P5) were all designed with vegetation and lifespans of 15 years (P3–1 to P3–14), 30 years (P2 and P5), 35 years (P1), and 50 years (P4–1, P4–2, P4–3) (Table S10 of SI). The environmental impacts per unit WQV, evaluated with TRACI, revealed variations in ecotoxicity impacts across the bioretention technologies (Fig. 3, Table S13 of SI). The P3–8 design exhibited the highest normalized ecotoxicity, followed by P3–14 (Xu and Zhang, 2019). P3–8 was designed with an internal water storage zone (IWSZ) of 60 cm, whereas P3–14′s IWSZ was 45 cm. Additionally, P3–14 utilized plant species of medium total nitrogen uptake capacity,

Table 3
Analyzed bioretention studies with their associated design details and descriptive statistics (note that the sample sizes N for the design basis variables differ by a maximum of five). Corresponding Paper ID for each reference is used to distinguish individual bioretention technologies for comparisons (e.g., in Figs. 3 and 4). Additional distinctions among scenarios (e.g., inclusion of an internal water storage zone, IWSZ) are further described in Tables S5, S10, and S15 of the SI. Acronyms and abbreviations include Imp. (imperviousness), WQV (water quality volume), and Std (standard deviation).

Paper ID	Imp. (%)	Design Storm (cm)	Drainage Area (m^2)	WQV (m^3)	System Area (m^2)	Reference
P1	100	5.1	44.1	2.1	–	(Vineyard et al., 2015)
P2	41	2.5	4900.0	52.5	405.0	(Flynn and Traver, 2013)
P3–1 to 4	75	20.3	2024.0	297.9	107.4	(Xu and Zhang, 2019)
P3–5,6	75	20.3	1336.0	196.6	71.2	(Xu and Zhang, 2019)
P3–7,8	75	20.3	1012.0	148.9	53.7	(Xu and Zhang, 2019)
P3–9 to 12	75	20.3	2024.0	297.9	107.4	(Xu and Zhang, 2019)
P3–13,14	75	20.3	1336.0	196.6	71.2	(Xu and Zhang, 2019)
P4–1	93	2.5	1169.0	25.9	40.0	(Bhatt et al., 2019)
P4–2	93	2.5	1394.0	30.9	62.0	(Bhatt et al., 2019)
P4–3	93	2.5	2709.0	60.1	72.0	(Bhatt et al., 2019)
P5	–	8.5	–	1.0	–	(Jeong et al., 2016)
P6–1	75	1.5	5300.0	35.0	69.0	(O'Sullivan et al., 2015)
P6–2	75	1.5	5300.0	35.0	41.4	(O'Sullivan et al., 2015)
P7–1	51	5.1	365,189.8	134.4	23.2	(Hengen et al., 2016)
P7–2	47	5.1	365,189.8	133.6	185.8	(Hengen et al., 2016)
P8	46.4	10.2	65,100.0	23,000.0	–	(C. C. Xu et al., 2021)
P9	100	–	7672.0	3027.0	–	(Xu et al., 2017)
P10	100	2.5	4047.0	97.7	25.0	(Bixler et al., 2019)
P11	80	2.5	4047.0	77.9	137.0	(Wang et al., 2013)
P12–1	92.5	1.5	800.0	10.6	49.3	(Mohamed et al., 2023)
P12–2	92.5	1.5	800.0	10.6	49.3	(Mohamed et al., 2023)
P12–3	92.5	1.5	800.0	10.6	49.3	(Mohamed et al., 2023)
P13	33.3	2.5	431,488.9	3775.5	–	(De Sousa et al., 2012)
P14–1	50	7.5	10,000.0	375.0	100.0	(Wang et al., 2016)
P14–2	50	11	10,000.0	550.0	100.0	(Wang et al., 2016)
P15–1 to 4	95	8	5000.0	75.0	200.0	(Andrew and Vesely, 2008)
P16–1 to 4	100	2.5	1590.0	38.4	91.0	(Sagrelius et al., 2023)
P16–5,6	100	2.5	530.0	12.8	30.0	(Sagrelius et al., 2023)
P16–7	100	2.5	1500.0	36.2	300.0	(Sagrelius et al., 2023)
P16–8	100	2.5	1500.0	36.2	368.0	(Sagrelius et al., 2023)
P16–9,10	100	2.5	2000.0	48.3	99.0	(Sagrelius et al., 2023)
P16–11	100	2.5	1400.0	33.8	52.0	(Sagrelius et al., 2023)
P16–12	100	2.5	3100.0	74.8	115.0	(Sagrelius et al., 2023)
N	32	32	32	33	28	
Median	92.5	2.5	2024.0	60.1	71.6	
Mean	80.5	6.4	40,947.0	998.1	109.8	
Std	21.2	6.6	114,150.4	4032.4	98.5	
Kurtosis	−0.6	0.7	7.2	30.1	3.4	
Skewness	−0.9	1.5	2.9	5.4	2.0	
Minimum	33.3	1.5	44.1	1.0	23.2	
Maximum	100	20.3	431,488.9	23,000.0	405.0	

namely *Salvia coccinea, Hymenocallis latifolia*, and *Coreopsis leavenworthii*. Although P3–14 achieved superior nutrient removal performance compared to other bioretention designs within P3, this benefit comes at the expense of increased environmental impacts (except eutrophication) and financial costs (Xu and Zhang, 2019). Xu and Zhang (2019) reported that incorporating an IWSZ and plants increases ecotoxicity, fossil fuel depletion, and global warming potential (GWP) impacts (Figure S2 of SI) as well as project costs. Increased project impacts and costs are primarily due to the additional materials, energy consumption, and extra maintenance needs, with electricity being the main contributor. In contrast, P4–1 recorded the lowest normalized ecotoxicity impacts (−1 in Fig. 3), representing earned ecotoxicity credits per unit WQV managed (Bhatt et al., 2019). Similarly, P2 demonstrated eutrophication and GWP credits per unit WQV managed (Flynn and Traver, 2013). The accumulated water quality benefits offered by biorientation treatment account for nearly all the avoided eutrophication, GWP, and ecotoxicity impacts (Bhatt et al., 2019; Flynn and Traver, 2013). Remarkably, P1 had the highest fossil fuel depletion per unit of WQV managed (+1 in Fig. 3) compared to other scenarios (Fig. 3, Figure S2 of SI), although the absolute impact was among the lowest (Vineyard et al., 2015) (Figure S3 of SI), thus highlighting the impact of the normalization process. The process of burning diesel fuel during transportation contributed to the fossil depletion impacts in P1 (Vineyard et al., 2015).

P12 scenarios (Mohamed et al., 2023) and P10 (Bixler et al., 2019) recorded relatively high impacts normalized to WQV for both climate change and fossil fuel depletion (above 0.5 in Fig. 3); therefore, the remaining studies using ReCiPe exhibited relatively lower climate change impacts (below 0.5 in Fig. 3) regardless of design lifespan or vegetation presence. Across the technologies reviewed, regardless of system area, the absolute impacts for fossil fuel depletion and climate change in P10 and P12 were influenced by the cumulative energy demand and greenhouse gas emissions during the construction phase, with P10 being more affected than P12. This is primarily due to the choice of construction materials and the use of diesel-powered machinery, e.g., trucks, loaders, and excavators (Bixler et al., 2019; Mohamed et al., 2023; Vineyard et al., 2015). Moreover, credits for GWP impacts from the removal of stormwater and air pollutants by the biofilter were reported by Mohamed et al. (2023). Therefore, the relatively high normalized impacts (except for climate change and fossil fuel depletion) per WQV observed in P10, P12, and other technologies may be largely influenced by the normalization process (Fig. 3, Figure S2). It should be noted that the P8 design had the highest absolute environmental impacts across all technologies (Figure S3). Furthermore, unlike other technologies, P8 (C. Xu et al., 2021) and the P12 scenarios (Mohamed et al., 2023) exhibited a declining trend in metal depletion (or mineral resource scarcity) per WQV managed, indicating a reduced consumption of key raw metals per m^3 of WQV managed. Typically, metal depletion is driven by raw materials, construction, disposal, and road transportation processes, which consume metal such as iron, copper, and magnesium and the energy required for their production (Mohamed et al., 2023; O'Sullivan et al., 2015; C. Xu et al., 2021). A spike in carcinogenic human toxicity per WQV was observed in P6 scenarios, whereby the associated absolute impacts were largely driven by emissions from transportation processes, reflecting a higher level of emissions toxic to human health from this process (O'Sullivan et al., 2015).

3.3.1. *Influence of vegetation on the sustainability of bioretention technologies*

Vegetation enhances water quality treatment through natural filtration and pollutant uptake, which offsets environmental impacts. Bioretention vegetation has to be drought-resistant and tolerant of saturated conditions to withstand the variability in watering associated with stormwater flow (Xu and Zhang, 2019). When comparing normalized impacts, some non-vegetated technologies (P6–1, P6–2, and P7–2; Fig. 3, Figure S2) showed an increase in metal depletion impacts; this could be attributed to the significant metal consumption in raw materials, transportation, construction, operation, and maintenance activities (O'Sullivan et al., 2015; C. Xu et al., 2021). Vegetated technologies (P10, P11, and P12 scenarios) showed lower absolute environmental impacts in some categories (Figure S3), particularly in pollutant-related impact metrics such as freshwater eutrophication and human toxicity (Bixler et al., 2019; Mohamed et al., 2023; Wang et al., 2013).

However, the gains from vegetation must be assessed in light of the WQV treated or managed. For instance, while P10 (vegetated and treats a large WQV) exhibits a high absolute freshwater eutrophication impact compared to P12 (also vegetated), its normalized impact (impact per unit WQV) is lower than that of the smaller P12 technologies. For technologies evaluated using ReCiPe, except for the system studied by P11 (Wang et al., 2013), technologies designed with vegetation had the highest impacts (normalized to WQV) on freshwater eutrophication, fossil depletion, ozone depletion, particulate matter formation, terrestrial acidification, human toxicity, and water depletion. Literature shows that bioretention technologies designed with vegetation offset annual ecotoxicity and eutrophication impacts due to their effectiveness in detaining and removing nutrients from stormwater runoff (e.g., P1, P5, P11) (Ali et al., 2021; Vineyard et al., 2015; Wang et al., 2013). Xu and Zhang (2019) found that while technologies with ground plants (vegetation) resulted in 51 % additional costs, they achieved a 54 % reduction in eutrophication (Fig. 3, Figure S2) with comparatively smaller increases in ecotoxicity (24 %–26 %) and fossil fuel depletion (28 %), when compared to the baseline scenario (P3–1, design without vegetation/plants). This indicates that vegetation contributes to the sustainability performance of bioretention technologies as they are beneficial for nutrient and hydrocarbon removal, soil bacteria quantity, plant survivability, and infiltration capacity. Vegetation also has the capacity for carbon sequestration through CO_2 uptake during growth, which was not considered in all studies. A review of GSI technologies based on published carbon footprint studies and field-measured carbon sequestration studies estimated that the carbon sequestration potential of bioretention systems may account for 70 % of their GWP produced during construction (Kavehei et al., 2018). Therefore, if primary measurement data were included in more studies, the normalized GWP impacts could approach zero or even negative, indicating a net benefit (e.g., as in Flynn and Traver, 2013).

Although this analysis focused on bioretention technologies as a case study due to their dense evidence base in the reviewed literature, this approach can be applied to other GSI types using the extracted data provided in the supplementary information Excel file and stormwater control design manuals in the literature. For example, in permeable pavements (no vegetation, storage/base layers), the required inputs include layer thickness, base gradation, and infiltration capacity; whereas for rainwater harvesting (capture/reuse), they include catchment area, storage volume, and drawdown conditions (Davis et al., 2022; WEF, 2012).

3.3.2. *Impact of imperviousness and water quality volume normalization on bioretention system sizing and sustainability performance*

Increased impervious areas, as well as design storms with lower annual exceedance probabilities, lead to substantial WQVs and a more pronounced first flush of pollutants and environmental impacts (Bhatt et al., 2019; Davis et al., 2022; Hengen et al., 2016; Wang et al., 2013; WEF, 2012). In one study, a sensitivity analysis of imperviousness (±10 %) revealed that higher imperviousness resulted in increased climate impacts by 0–18 %, while reduced imperviousness lowered these impacts by 0–12 % (Wang et al., 2013). Increased WQV and system size can enhance the environmental benefits of bioretention technologies if the additional stormwater remains within the system's design capacity (Vineyard et al., 2015). Exceeding it leads to overflow and potential damage to plant life and reduced functionality of the system, which in turn raises maintenance costs and environmental impacts (Vineyard

et al., 2015).

As an example of the importance of these relationships, the P8 bioretention exhibited the highest absolute impact for freshwater eutrophication at 62.32 kg P eq, followed by P10 at 2.77 kg P eq, compared to much lower impacts in P11 (0.013 kg P eq) and intermediate impacts in the P12 scenarios (ranging from 0.382 to 0.637 kg P eq). However, when these values were normalized by their respective WQVs (Table 3), the P12 scenarios showed higher normalized freshwater eutrophication impacts (0.0602, 0.0452, and 0.0361 kg P eq/m^3) than P8 (0.00271 kg P eq/m^3) and P10 (0.0283 kg P eq/m^3), with P11 remaining very low (0.000161 kg P eq/m^3) (Fig. 3, Table S8 of SI). This analysis reveals that while technologies with larger WQV (such as P8, P9, P10, and P11) may exhibit higher absolute environmental impacts simply due to their size, their performance in terms of impact per unit of water quality volume treated is generally more efficient than that of smaller technologies (like the P12 scenarios), though studies may vary by system boundaries (Fig. 4). This trend is observed consistently across the impact categories studied, highlighting the importance of normalizing impacts to system capacity when evaluating environmental performance. Absolute environmental impacts can be high for larger technologies compared to smaller technologies simply because large technologies manage more stormwater, but normalization to WQV provides a more equitable basis for comparison.

3.3.3. *Influence of bioretention design choices, filter media, amendments, and design lifespan*

GSI design choices (including system components, design lifespan, filter media composition, and material selection) influence the performance, environmental, and economic impacts of these technologies. For example, P3 scenarios were designed with sand-dominated filter media, IWSZs of varying depths, and different plant species (Xu and Zhang, 2019), while P4 scenarios also featured sand-dominated filter media and plants, but lacked the IWSZ and instead used additional proprietary treatment technologies (i.e., concreted-housed ceramic-like aluminum oxide sorptive media) (Bhatt et al., 2019). As a result, P4 scenarios achieved more avoided environmental impacts compared to P3 scenarios due to the reduction of multiple influent pollutant loads (Bhatt et al., 2019).

Furthermore, the amendment of sorptive soil media (e.g., SBAC – sludge-based activated carbon, CAC – commercial activated carbon) has been reported to increase bioretention system lifespan, pollutant removal, overall environmental impacts, and monetized benefits from extended lifespans (Mohamed et al., 2023). P12–1, P12–2, and P12–3 had the highest normalized environmental impacts for freshwater eutrophication, fossil depletion, ozone depletion, particulate matter formation, terrestrial acidification, water depletion, and carcinogenic human toxicity (Mohamed et al., 2023). These technologies considered the same vegetation type in their design but with different assumed lifespans within 100 years of service, as well as variable filter media compositions (i.e., baseline - sand layer only for P12–1 at 30 years lifespan, with SBAC or CAC at 40 years for P12–2, with SBAC or CAC at 50 years for P12–3) (Mohamed et al., 2023). The larger environmental impacts from P12–1 can be attributed to the amendment of sorbents (SBAC, CAC) into the filter media. Additionally, Mohamed et al. (2023) reported improved environmental impacts by up to 29 % when a 40-year lifespan was assumed (i.e., P12–2) and up to 40 % with a 50-year lifespan (i.e., P12–3). Longer lifespans decrease the need to reconstruct the bioretention technologies, thus resulting in lower environmental impacts and economic costs (Mohamed et al., 2023; Vineyard et al., 2015). As a trade-off, the addition of activated carbon sorbents increased total bioretention project costs; however, the net monetized benefits achieved from the extended lifespan outweighed the additional costs (Mohamed et al., 2023).

3.3.4. *Influence of system boundary choices on sustainability impacts*

While 13 bioretention studies (Table 3) included raw materials and transportation (associated with raw materials and construction), and all 16 studies included the construction and maintenance phases in their

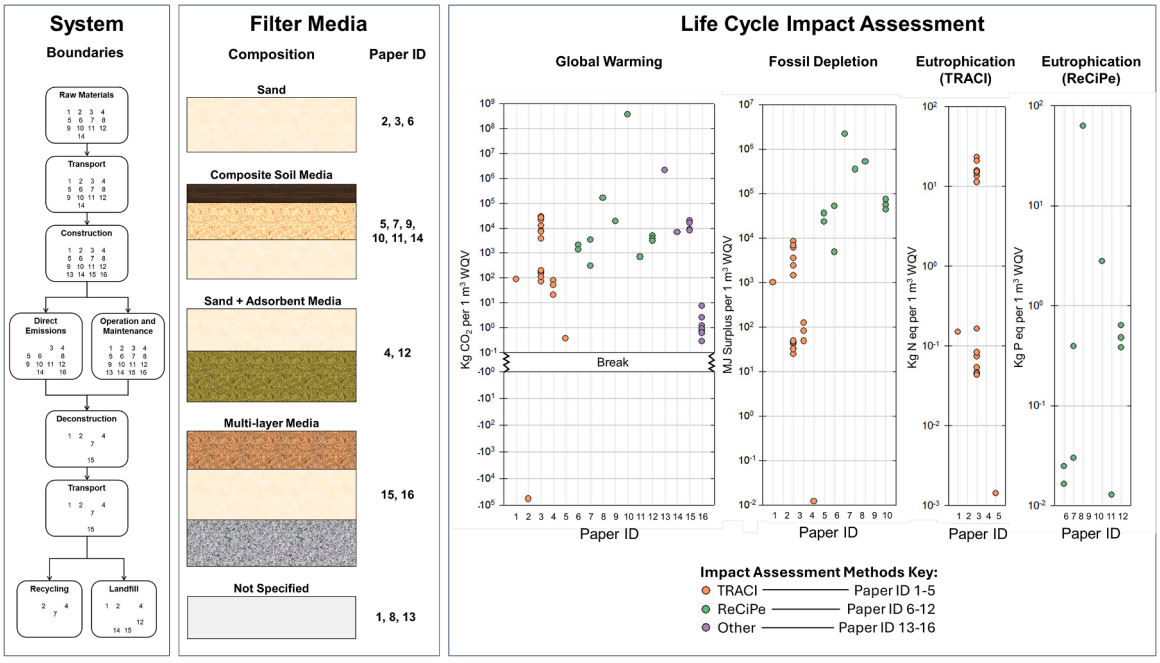


Fig. 4. Influence of LCA system boundary choices and bioretention filter media composition on environmental impacts. The first panel (system boundaries) shows all the life cycle phases and the corresponding paper IDs (Table 3) that considered them in the system boundary. The second panel (filter media) indicates the baseline filter media composition and corresponding paper IDs (Tables S5, S10, S15 of SI). Composite soil media includes a mix of sand, loam, fines (silt and clay) and organic matter. Adsorbents include sludge-based activated carbon and commercial activated carbon, aluminum oxide, and hollow-fiber membrane. The third panel (impact assessment) provides a scatter plot of selected impact categories based on complete data availability (normalized to WQV) for TRACI, ReCiPe, and other LCIA methods (i.e., CML, ILCD, IPCC, and EIO-LCA) and the corresponding paper IDs. Eutrophication impacts are divided by LCIA method due to different units. Note that the extra dots for a given paper ID indicate the environmental impacts of different bioretention design scenarios within the same paper.

LCA system boundaries (Fig. 4), fewer studies extended their boundaries to other life cycle phases. Eleven studies accounted for direct emissions, 5 considered deconstruction, 5 included additional transportation, 3 included recycling or reuse, and 6 extended their boundaries to cover the landfill or disposal phase (Fig. 4). The variation in system boundary coverage can lead to differing environmental and economic impact results, since including more phases often captures additional emissions and resource use. Studies that included a higher number of life cycle phases (e.g., P7) showed higher impacts normalized to WQV, whereas studies like P16, which focused only on construction, operation, and maintenance, reported relatively low normalized impacts.

While some studies concentrated on a combination of sand, natural soil, loam, and compost as filter media, sand was the most utilized. Filter media choice directly influences pollutant removal capacity, nutrient release, and the embodied environmental impacts of material production and construction processes. Compost-based media and the emergence of byproduct media, for instance, are low-cost and contain natural microbial communities capable of biodegrading several stormwater pollutants (Esfandiar et al., 2022; Hort et al., 2009; James and Yadav, 2021) but they may also introduce new resource demands.

The construction phase accounted for up to 98 % of financial costs and 90 % of environmental impacts in P1 (Vineyard et al., 2015), and 77 to 99 % of environmental impacts in P12 scenarios (Mohamed et al., 2023). Major contributors included the raw materials (e.g., sand, gravel) and the transportation of the raw materials to the construction site (Flynn and Traver, 2013; Mohamed et al., 2023). In two studies, burning diesel fuel was the largest contributor to most environmental impact categories and maintenance costs, however, regular maintenance (e.g., excavating and restoring filter media layers) is critical to bioretention performance (Liu et al., 2014; Vineyard et al., 2015), highlighting the need to balance functional performance, costs, and emissions.

3.3.5. *A trade-off between life cycle environmental costs and impacts*

Within the bioretention analysis, 37 bioretention design scenarios with both LCA and LCC results (i.e., P1, P2, P3–1 to P3–14, P9, P10, P14–1, P14–2, P12–1 to P12–3, and P16–1 to P6–12), were analyzed to elucidate the relationship between environmental impacts (GWP per unit WQV), and economic impacts (net present value costs, NPV per unit WQV) (Fig. 5, additional details in Section S4 of SI). GWP was selected because it was the most widely reported environmental impact category. For a fair cost comparison, all reported costs were converted into NPV costs (Table S16, Equations S3-S5). This conversion incorporated the individual design lifespans and discount rates (the rates used to discount future cash flows to present value) provided in each study. For studies that did not specify a discount rate, a default value of 5 % was adopted, as it is the most common rate in the reviewed literature (Table 2; Section S4 of SI).

The interquartile range for GWP per WQV ranged from 0.066 to 74.85 kg CO_2 eq/m^3 (median of 1.00 kg CO_2 eq/m^3; Fig. 5) compared to the interquartile range for NPV per WQV which ranged from 135.74 to 2573.02 \$/m^3 (median of 620.31 \$/m^3), indicating variation in results among the various design scenarios. Following the trade-off frontier (red line, Fig. 5), the bioretention designs of P2, P3–1, and P9 (Table 3, Section S3 of SI) provided the most favorable trade-offs between economic impact (NPV per WQV) and environmental impact (GWP per WQV). For instance, decision-makers may prioritize designing the

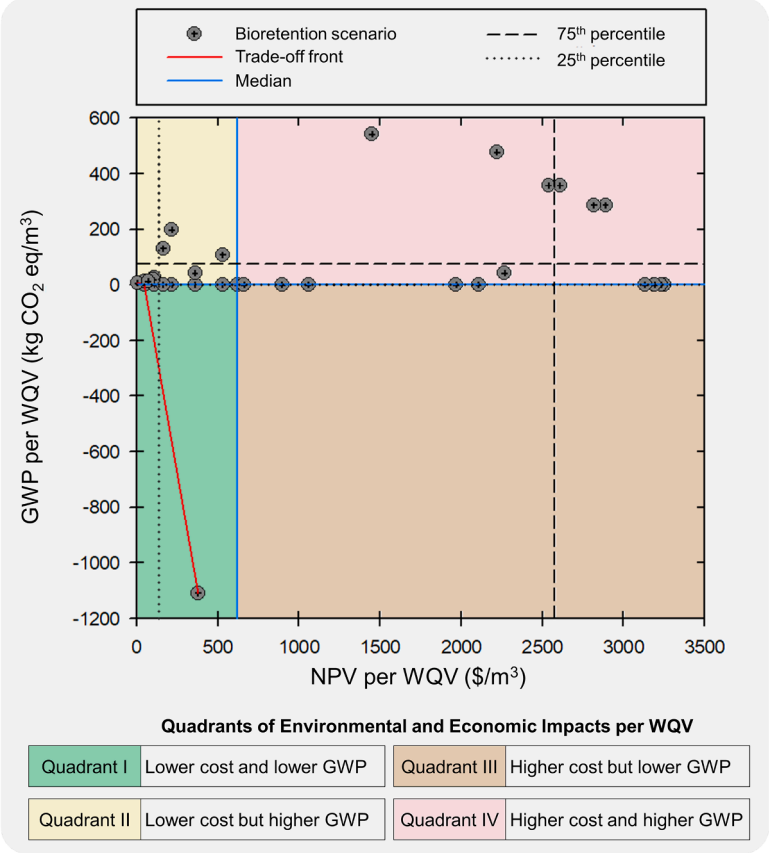


Fig. 5. A scatter plot of 37 bioretention design scenarios (Figure S4, Table S16 of SI) showing the trade-off between global warming potential (GWP) impacts per water quality volume (WQV) (kg CO_2 eq/m^3) on the y-axis and net present value (NPV) per WQV (\$/m^3) on the x-axis. Vertical and horizontal median blue lines divide the plot into four colored quadrants (I to IV). Dotted and dashed lines represent the 25th and 75th percentiles, respectively. The distance between them is the interquartile range, representing the spread of the middle 50 % of the designs. The trade-off frontier (red line) connects the set of designs (P2, P3–1, and P9) that offer the best balance between environmental and economic impacts.

lower-cost end of the frontier while allowing for a certain amount of GWP if cost reduction is crucial (P9). On the other hand, even if the cost is higher, they can choose designs on the border with the lowest GWP (or GWP credits) if reducing environmental impacts is of the utmost importance (P2). Although P2, P3–1, and P9 are most favorable in terms of the cost–GWP trade-off, they may not deliver all potential benefits or meet other competing goals; therefore, decision-makers must balance these additional trade-offs when choosing a design. The significance of normalizing impacts on a per-unit basis was further emphasized; for instance, P12 provides alternatives that result in GWP credits and benefits compared to other designs (such as P6 and P9); however, these benefits come with higher overall costs and GWP per WQV treated (Mohamed et al., 2023). Finally, studies vary in system boundaries, which may influence comparative results (Fig. 4).

3.4. Charting a path forward for LCA and LCC to inform GSI decision-making and design

3.4.1. Considering the context of existing policy and regulatory frameworks

LCA and LCC analyses can enhance GSI design, implementation, and alignment with local policy and regulations (Fig. 6). As cities compete with urbanization, population growth, and climate change, GSI has been increasingly integrated into long-term planning and policy (Bashirpour Bonab et al., 2023; J. Chen et al., 2023; Dhakal and Chevalier, 2017; Javed et al., 2021). The adoption, design, implementation, and operation of GSI have been historically driven by climate, regulatory policies and pressure, social-psychological, and socio-economic factors which vary based on the region (Choat et al., 2023; Drescher and Sinasac, 2021; McPhillips and Matsler, 2018; Qi and Barclay, 2021; Rieck et al., 2022). Regional variability in policy translates directly into variability in GSI adoption and practices, affecting sizing, design components, expected lifespan, and which performance metrics are monitored. This may help explain why GSI LCA/LCC research is geographically dominated by China and the USA (Figure S1 and Table S4 of SI). As context, a review of municipal green infrastructure plans across 24 Scandinavian municipalities (Norway, Sweden, Denmark) by Nordh and Olafsson (2021) found that all municipalities had green infrastructure strategies. However, only 60 % had a dedicated green infrastructure or green space plan, thus highlighting policy heterogeneity and the distinction between green infrastructure policy and GSI implementation (Nordh and Olafsson, 2021). In the global south, urban cities' prevailing policies and governance dimensions are not structured to fuel GSI documentation and implementation despite the environmental, economic, and social vulnerability to stormwater flood impacts and the need for GSI technologies (Ibrahim et al., 2023). Together, these regional differences influence where particular GSI practices are applicable and their performance, therefore underscoring the need for context-specific GSI LCA/LCC analyses. Locally, many U.S. municipalities rely solely on the municipal separate storm sewer system (MS4) permit requirements. Under the MS4 Permit of the U.S. EPA's Clean Water Act – specifically Minimum Control Measure 5: "Management of Post-Construction Site Runoff" – municipal operators must control stormwater discharges from developed and redeveloped sites (USEPA, 2003). This requirement has driven widespread adoption of GSI, as many municipalities earn water-volume reduction credits from GSI practices (Vogel and Moore, 2016). However, once GSI design parameters have been reviewed or approved and the infrastructure installed, there is no formal requirement for commissioning, recommissioning, or decommissioning, limiting the ability to quantify long-term sustainability impacts. Furthermore, because MS4 permits are administered and overseen at the state level, regulations and enforcement vary widely. For example, some design guidance may reference designing to a certain storm (e.g., 80 % annual exceedance probability or 2-year, 24-hour event), while another may require designing to a water quality runoff volume (e.g., capturing 1 in design storm). In many cases, stormwater management suggests designing to "maximum extent practicable," resulting in no uniform design guidance (Copeland, 2016). Integrating LCA and LCC tools can help fill such gaps and strengthen existing mandates by providing a holistic framework for GSI design and assessment; however, these tools would also require a policy shift towards the adoption of a standardized life cycle inventory (emission flows and costs) and FU for each GSI technology.

3.4.2. Moving toward holistic sustainability assessments of GSI

While countless studies exist on water quality improvements or volume reductions, very few LCA and LCC papers incorporate these metrics as a result. Standardized performance data could be incorporated into LCA or LCC workflows to better quantify sustainability impacts (Fig. 6). For example, a bioretention technology designed with vegetation and an IWSZ showed improved water quality performance (i.e., nutrient reduction) but also led to increased life cycle costs and greater environmental impacts (except eutrophication) (Xu and Zhang, 2019). LCA could also enhance the selection of other design parameters within bioretention; for example, materials (e.g., sand, fly ash, or wood

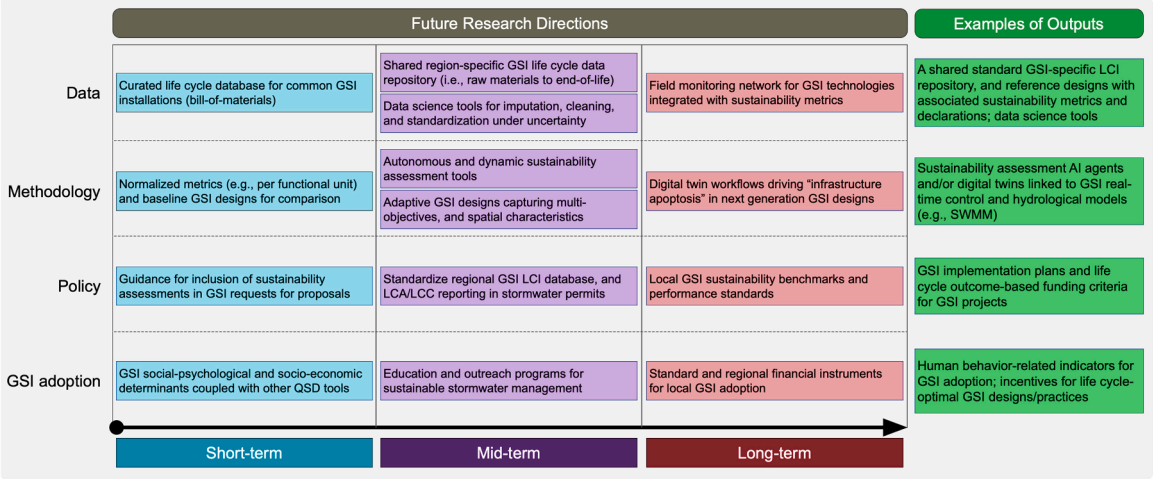


Fig. 6. Summary roadmap for advancing the sustainability of green stormwater infrastructure (GSI). The horizontal axis shows the estimated timeline (for example, depending on context: 0–2 years for short-term, 2–5 years for mid-term, and 5+ for long-term). The vertical axis groups research focus areas or challenges by category (data, methodology, policy, and GSI adoption). The green panel on the right lists example outputs for each category. Note that the directions in the figure are not in a specific hierarchy, and research directions may overlap in categories and be iterative in nature. Abbreviations: LCI = life cycle inventory, QSD = quantitative sustainable design, SWMM = stormwater management model.

chips) used to target specific water quality pollutants may be sourced locally as byproducts, thus decreasing environmental impacts associated with transportation or production, respectively. Media, such as coconut coir, may be effective for pollutant removal and stabilization (Esfandiar et al., 2022; James and Yadav, 2021; Romali et al., 2023; Tota-Maharaj and Cheddie, 2015); however, the transportation needed to the U.S., for example, is likely to impact overall sustainability. These trade-offs highlight the challenge of enhancing one environmental goal (e.g., eutrophication) without sacrificing others (e.g., acidification). In these cases, LCA can be used to identify alternatives or compromises that best align with stakeholder goals.

Furthermore, additional "ecosystem services" or co-benefits of GSI (e.g., groundwater recharge, heat island mitigation, improved habitat, and carbon sequestration) are not easily incorporated into LCA and LCC (Zhang et al., 2010b, 2010a). While their valuation has been difficult to assess, some incentive programs exist for these benefits (Dhakal and Chevalier, 2017). These programs are variable, with some programs partially or fully covering the stormwater utility fee through discounts and others providing rebates to offset installation and maintenance or awarding grants to install GSI. If fully realized, incentive programs can be beneficial to private property owners and public utilities. LCA and LCC can play a critical role in aligning fee assessment and savings projections with performance metrics for incentive programs. LCA and LCC also provide an opportunity to align the FU with specific goals (e.g., ecosystem services) to better inform incentive schemas (Teston et al., 2024). One challenge to the implementation of LCA/LCC for incentive programs is understanding system longevity, maintenance requirements, and EoL; more studies are needed to provide such data.

Finally, quantitative sustainable design (QSD) – a methodology to inform holistic GSI technology design and operational decisions amidst uncertainty – can provide a framework for simultaneously considering these many design considerations and potential co-benefits. In summary, QSD encompasses three primary steps executed under uncertainty: (i) defining the problem space, (ii) establishing system design simulation algorithms, and (iii) characterizing system sustainability by assessing environmental (e.g., using LCA), economic (e.g., using LCC or techno-economic analyses), human health (e.g., using QMRA), and social indicators (e.g., using S-LCA) (Li et al., 2022). While some reviewed studies have made significant progress by integrating "socioecological influences" (primarily environmental impacts with social consequences), future sustainability assessments could better capture the social dimension and trade-offs among competing design objectives of GSI and/or hybrid stormwater infrastructure (green-gray) by coupling multi-objective optimization with LCA, LCC, QMRA, and S-LCA indicators or qualitative social criteria from the QSD framework, in line with international guidance (Benoît and Mazijn, 2009; Benoît Norris et al., 2020; Li et al., 2022; Liu et al., 2021; Y. Zhang et al., 2025; Zhu et al., 2023). Along with uncertainty analysis (e.g., using Monte Carlo simulations), sensitivity analysis and/or scenario analyses in the QSD framework can then reveal the contributions of individual uncertain parameters and test alternative design or operational strategies (Li et al., 2022). Lastly, multicriteria decision analysis can integrate quantified sustainability indicators and stakeholder priorities to inform decision-making (Ebrahimian and Wadzuk, 2022; Li et al., 2022).

3.4.3. *Integrating artificial intelligence (AI) to advance LCA and LCC analyses, and future GSI designs*

Despite their promise, sustainable design tools often remain static. LCA and LCC analyses cannot respond to system changes and are often limited by the absence of detailed site-specific inventory data (Martínez-Ramón et al., 2024). The lack of data encourages the use of proxy or assumed values, which increases uncertainty (Song, 2019). Although only a few studies conduct both uncertainty and sensitivity analyses (Table 2), the gaps in real data ultimately compromise the accuracy and quality of sustainability analyses. In addition, manual selection of inventory data, processes, and execution of analyses can be laborious and prone to human error. The fast-paced rise of artificial intelligence (AI) and real-time control (RTC) for automated hydraulic control and seamless data collection and management offers a path forward, despite barriers such as high costs, maintenance demands, and regulatory compliance (Cornago et al., 2023; Naughton et al., 2021). For instance, RTC strategies in GSI dictate operational conditions, and as a result, they have been shown to: (i) enhance bioretention performance by improving pollutant removal (Persaud et al., 2019; Shen et al., 2020; W.D. Xu et al., 2021; Zhang et al., 2024); (ii) boost flood control efficacy (Bilodeau et al., 2018; Li and Burian, 2023; Persaud et al., 2019; Poresky et al., 2015); (iii) reduce uncontrolled overflows in rainwater harvesting systems (Persaud et al., 2019); and, (iv) increase water retention in green roofs (Brasil et al., 2021; T. Chen et al., 2023). Meanwhile, AI algorithms coupled with stormwater modeling tools (e.g., EPA Storm Water Management Model – SWMM) can optimize GSI design, operational planning, performance monitoring, and maintenance scheduling by analyzing data from RTC sensors, rainfall patterns, soil and land use data, and hydrological parameters (Essamlali et al., 2024; Ramovha et al., 2024).

At the same time, AI-driven LCA and related tools are becoming more precise, scalable, and dynamic (Hosamo et al., 2024; Vital et al., 2024). For example, machine-learning approaches can streamline inventory data collection and boost accuracy in tools like LCA and LCC through approaches such as: (i) neural-network clustering and K-nearest neighbors to extract and clean data (Ibn-Mohammed et al., 2023; Köck et al., 2023; Romeiko et al., 2024; Wehner et al., 2022); (ii) support-vector machines and deep neural networks to correct errors and cluster inventory records (Ibn-Mohammed et al., 2023; Yılmaz and Şahin, 2023); (iii) generative adversarial imputation nets and artificial neural networks for addressing data gaps (Ibn-Mohammed et al., 2023; Qin et al., 2024; Song et al., 2017; W. Wang et al., 2022; Yoon et al., 2018); and (iv) large language models to support workflows such as recommending emission factors (Balaji et al., 2025; Cornago et al., 2023). Furthermore, multimodal AI agents have the potential to reinvent and automate the conventional practice of conducting sustainability assessments and controlling stormwater management technologies (Song et al., 2024; Z. Zhang et al., 2025). Broader research and development of these AI-assisted tools could advance the speed and accuracy of GSI LCAs and LCCs.

Looking toward "cities of the future" (Bashirpour Bonab et al., 2023; Javed et al., 2021), we envisage that advances in AI- and RTC-enabled digital twin workflows with GSI and QSD tools (e.g., LCA, LCC) (Brasil et al., 2022; He et al., 2025; Kim et al., 2025; Neupane et al., 2025; Petri et al., 2025) will advance QSD research and adaptive GSI designs. This integration supports future GSI technologies that can autonomously manage the planned, automated winding down of GSI systems or components at their end-of-life. End-of-life could be defined as when GSI fails to meet predetermined performance metrics, resilience thresholds, or sustainability benchmarks. In practice, these systems could (i) continuously monitor performance via RTC and AI, (ii) trigger decommissioning or retrofit signals once predefined indicators cross thresholds, (iii) virtually assign infrastructure components to disposal streams, and (iv) update digital twins in real-time. These future directions can help to overcome current barriers presented in this study, while ensuring that modern GSI technologies become adaptive, efficient, and sustainable over their entire life cycle (Fig. 6).

4. Conclusion

This systematic review of literature over the past two decades characterized the sustainability of green stormwater infrastructure (GSI) using life cycle assessment (LCA) and life cycle costing (LCC). Our findings suggest that effective GSI design should prioritize and balance (1) sufficient system size and water quality volume to attenuate flow and reduce normalized sustainability impacts, (2) the inclusion of vegetation to enhance pollutant removal, and (3) extended lifespans to offset the

initial environmental and economic impacts. Such a balanced approach can lead to technologies that not only achieve water quality targets but also minimize costs and environmental burdens across multiple impact categories. The study also reinforces the need to evaluate GSI technologies and life cycle environmental and cost impacts on a per-unit normalized basis (e.g., per m^3 of WQV treated) rather than absolute terms, ensuring that comparisons account for differences in scale, design, and functionality. The trade-off analysis revealed that a GSI design with a low economic impact does not automatically equate to a low environmental impact per unit of WQV treated. Therefore, decision-makers must weigh the importance of minimizing costs against achieving lower environmental impacts using LCC and LCA, among other decision-making criteria. To reduce variability arising from regional policy and database differences, we recommend a shift toward a standardized life cycle inventory and impacts expressed in a common functional unit. Looking ahead, new developments and research in real-time control (RTC) and AI-driven decision support present viable ways to get around existing drawbacks, such as incomplete or faulty inventory data and the time-consuming nature of data collection and assembly, thus improving LCA and LCC workflows. Collectively, these recommendations and advances can support the use of LCA and LCC to guide decision-making for sustainable GSI systems to improve stormwater management and address environmental challenges with GSI.

CRediT authorship contribution statement

Isaac Oluk: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Sophia Del Rey:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Data curation. **Mary E. Foltz:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Funding acquisition, Conceptualization. **Jaime C. Schussler:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Funding acquisition, Conceptualization. **Diana M. Byrne:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.watres.2025.124586.

Data availability

All data are available in the supplemental information (SI) files.

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